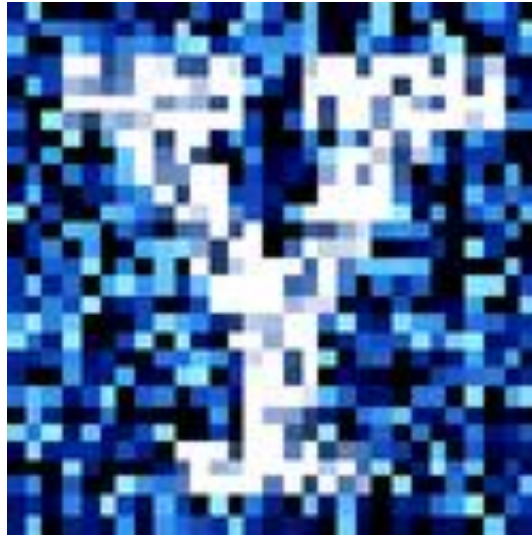


YData: Introduction to Data Science



Class 18: Hypothesis test for proportions

Overview

Review of Statistical inference

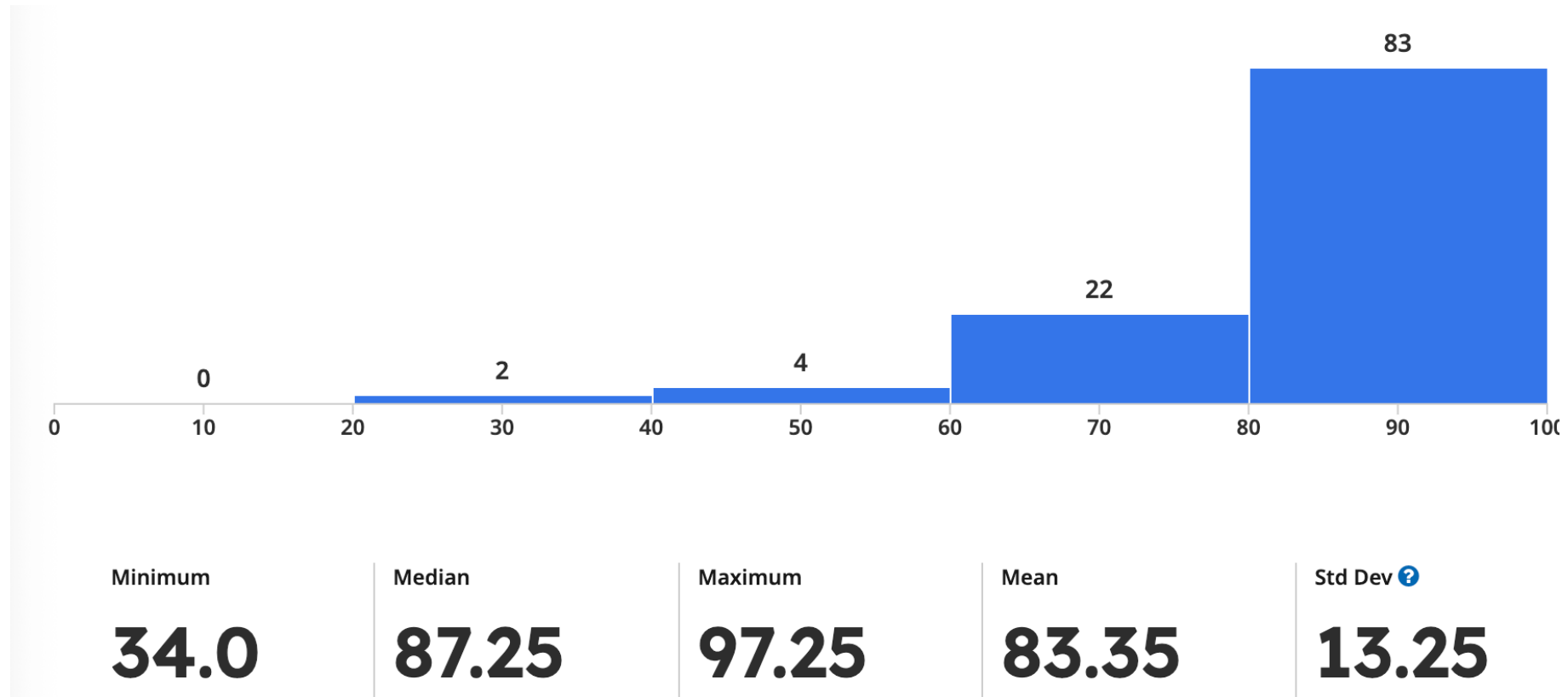
Hypothesis tests

- Hypothesis tests for a single proportion



"Null Hypothesis" [\[Long Description\]](#).

Midterm!



Please submit your regrade request on gradescope **in a week!**

Reminder

A **polished** draft of the project is due on **April 10th**

Homework 7 is due on **Sunday March 29th**

Also, homework 8 is posted and will be due on **Sunday April 5th**

Statistical Inference

Inference

Population: all individuals/objects of interest

- E.g., all voters

A parameter is number associated with the population

- E.g., The proportion of all voters who voted for Biden: π_{Trump}

Sample: A subset of the population

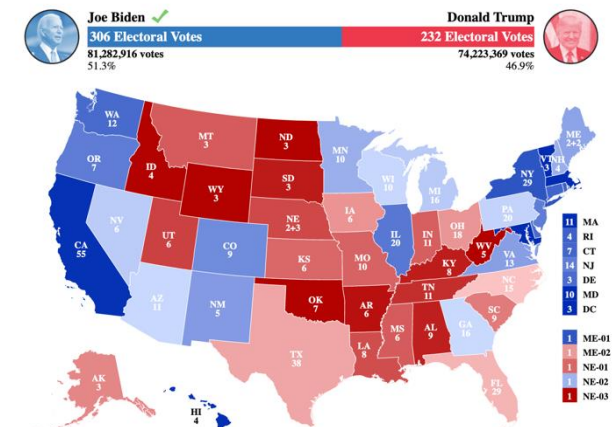
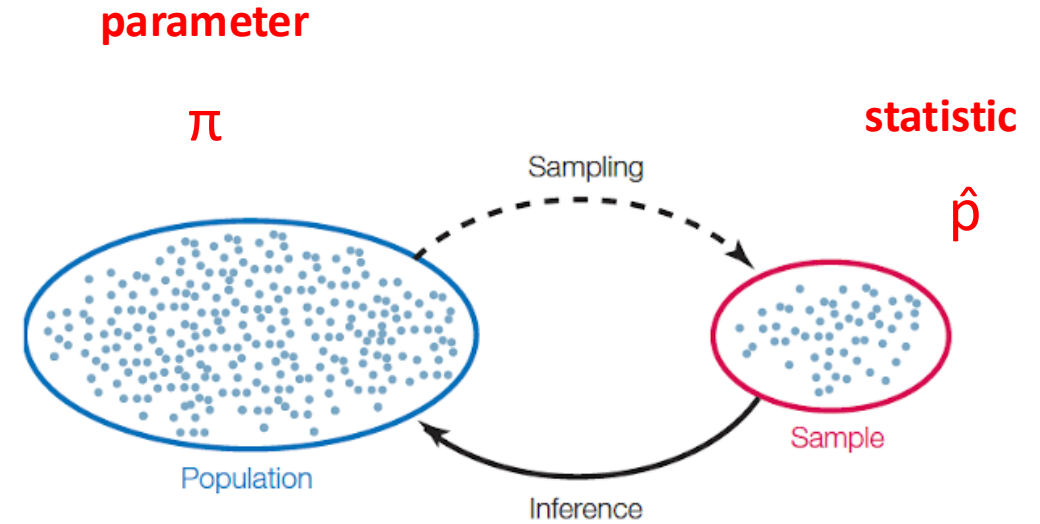
- E.g., 1000 randomly sampled voters

A statistic is number calculated from the sample

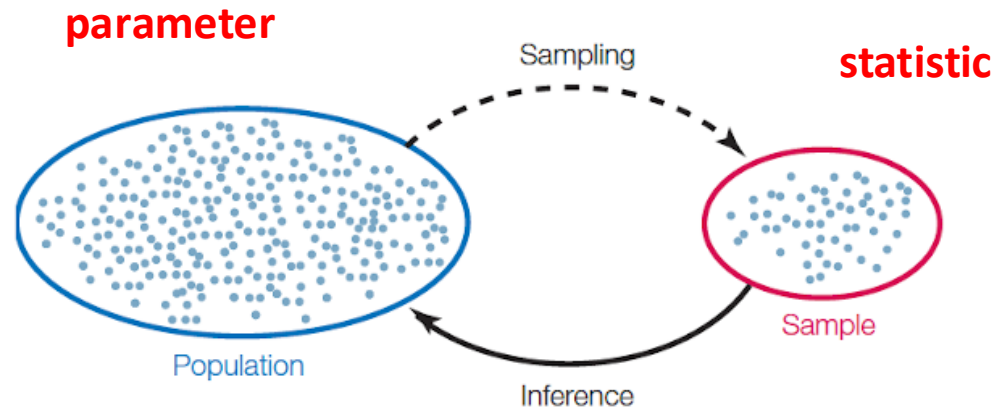
- e.g., The proportion in the sample who voted for Biden: $\hat{p}_{\text{Trump}}$

Statistical Inference: Making conclusions about a population based on data in a sample

- E.g., using $\hat{p}_{\text{Trump}}$ as an estimate of π_{Trump}



Examples of parameters and statistics



	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Proportion	π	$\hat{p}$
Standard deviation	σ	s
Correlation	ρ	r

Probability distribution of a statistic

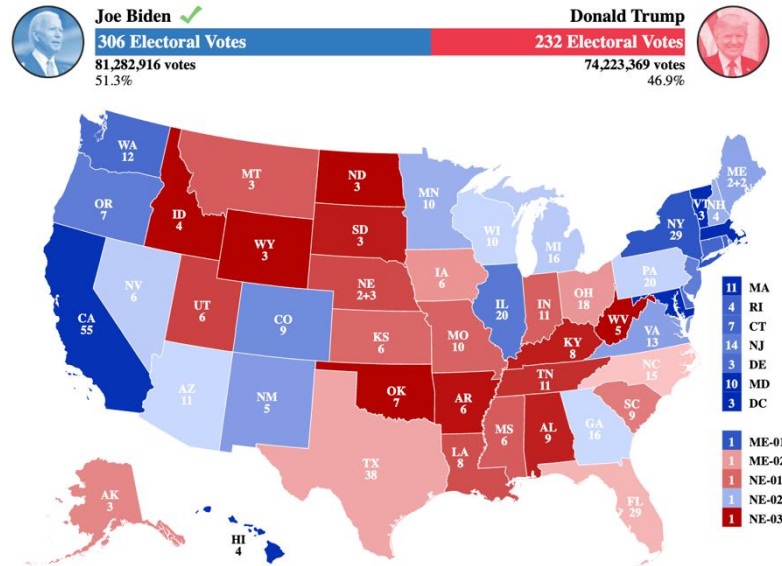
Values of a statistic vary because random samples vary

A **sampling distribution** is a probability distribution of *statistics*

- All possible values of the statistic and all the corresponding probabilities
- We can approximate a sampling distribution by a simulated statistics

π_{Trump}

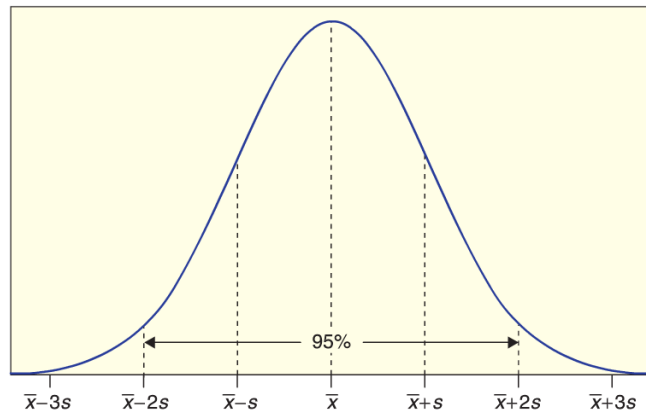
$n = 1,000$



$\hat{p}_{\text{Trump}}$



$\hat{p}_{\text{Trump}}$



Sampling distribution!



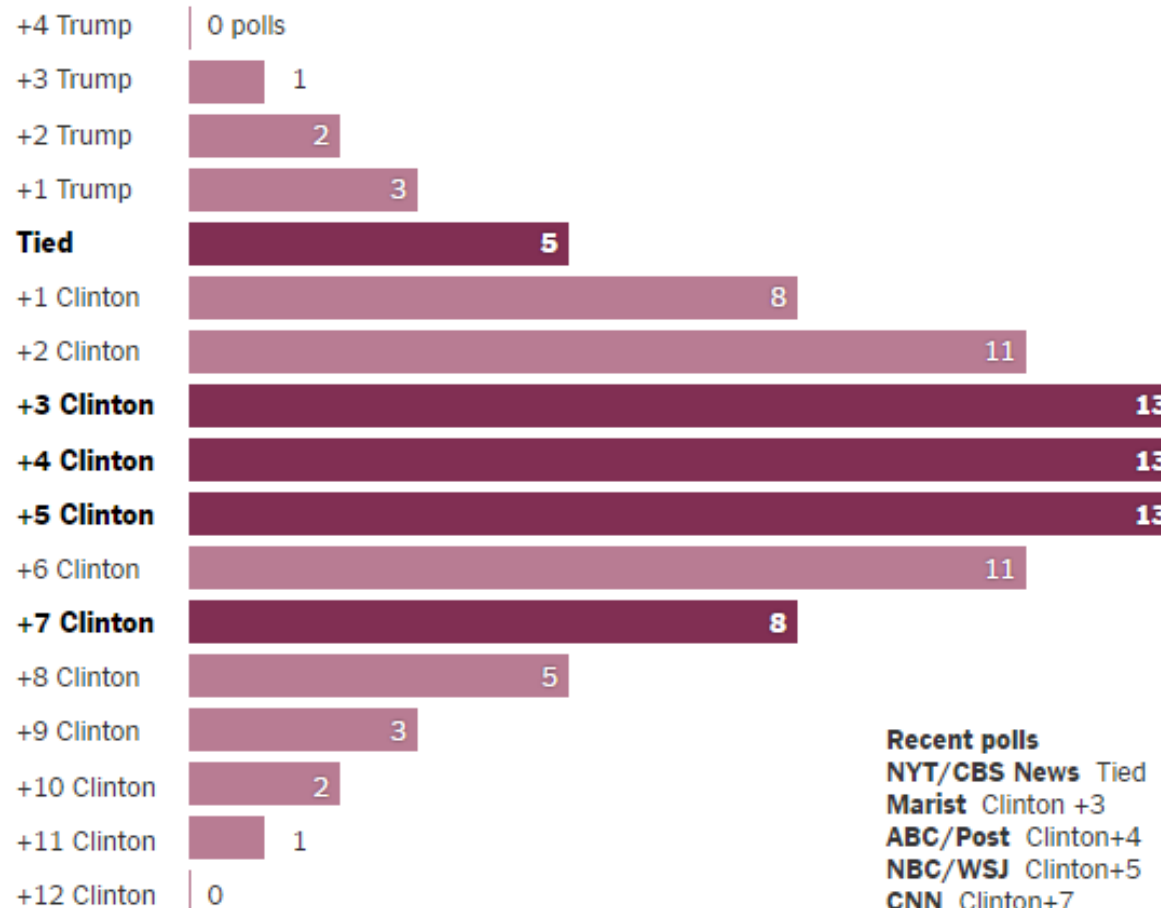
$\hat{p}_{\text{Trump}}$

Confused by Contradictory Polls? Take a Step Back

Noisy Polls Are to Be Expected

If Hillary Clinton were up by a modest margin, there would be plenty of polls showing a very close race — or even a Trump lead.

A simulation of 100 surveys, if Mrs. Clinton were really up 4 points nationally.



Sampling distribution of $\hat{p}_{\text{Clinton}}$



What parameter are they trying to estimate?

Questions?

Hypothesis tests

Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population (parameter)

Example 1: The average body temperature of humans is 98.6°

How can we write this using symbols?

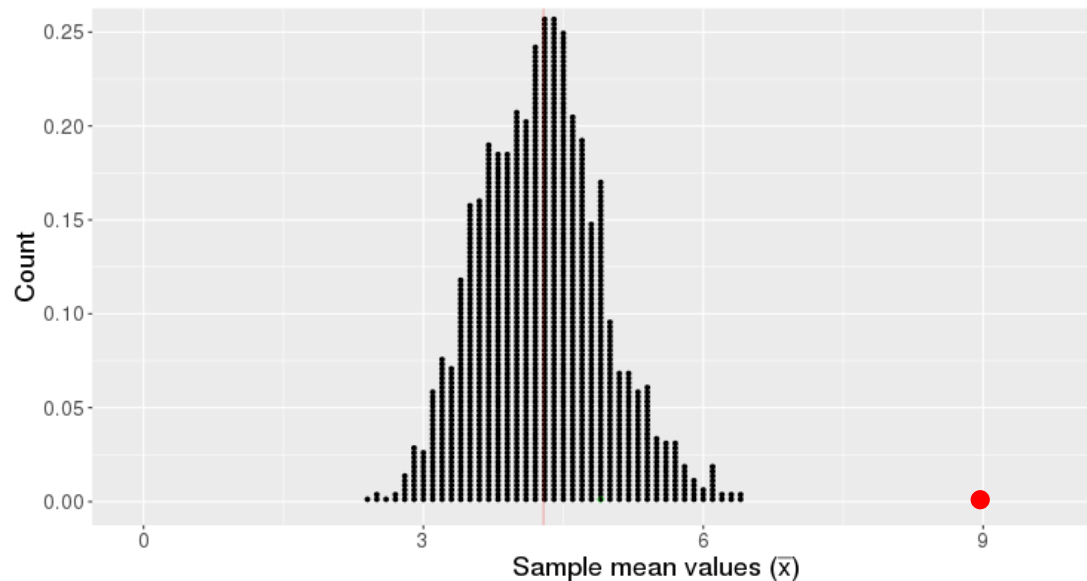
- $\mu = 98.6$

Basic hypothesis test logic

We start with a claim about a population parameter

- E.g., $\mu = 4$ 

This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

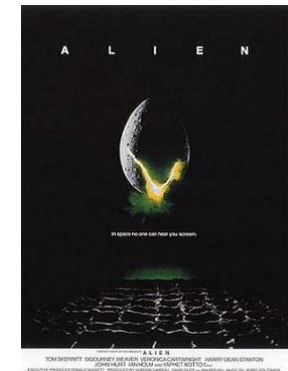
Motivating example: The Bechdel Test

Question: Do less than 50% of movies pass the Bechdel test?



Questions:

- What is the population/process?
- What is our parameter of interest?
 - What symbol should we use to denote it?
- What is our statistic of interest?
 - What symbol should we use to denote it?

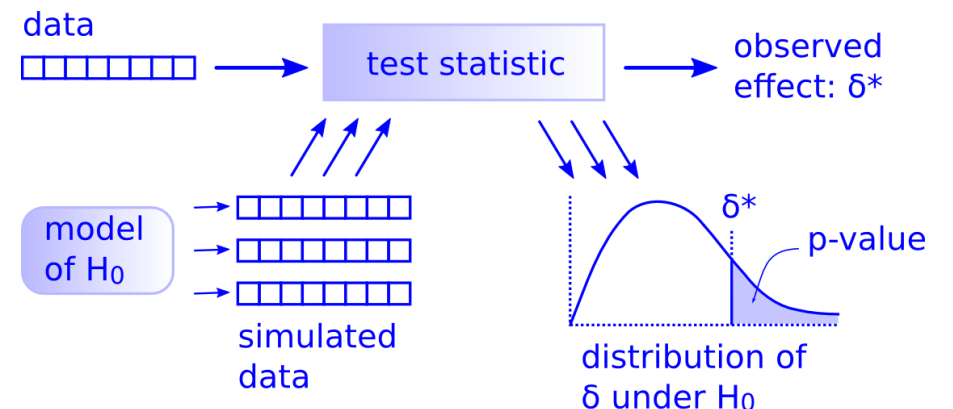
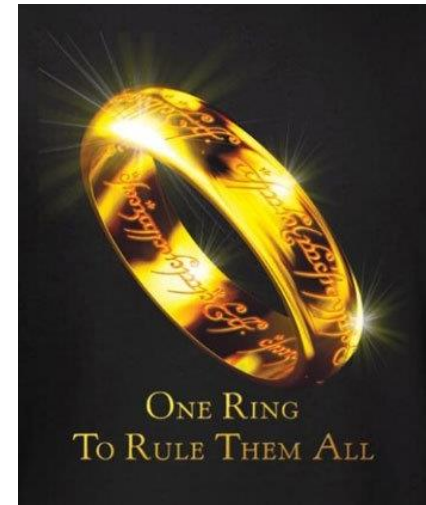


Steps needed to run a hypothesis test

To run a hypothesis test, we can use 5 steps:

1. State the null and alternative hypothesis
2. Calculate the observed statistic of interest
3. Create the null distribution
4. Calculate the p-value
5. Make a decision

Let's go through these steps now...



Do less than 50% of movies pass the Bechdel test?

Step 1: state the null and alternative hypotheses

If only 50% of the movies passed the Bechdel test, what would we expect the value of the parameter to be?

$$H_0: \pi = 0.5$$

If fewer than 50% of movies passed the Bechdel test, what would we expect the value of the parameter to be?

$$H_A: \pi < 0.5$$

Observed statistic value

Step 2: calculate the observed statistic

There are 1794 movies in our data set

Of these, 803 passed the Bechdel test

What is our observed statistic value and what symbol should we use to denote this value?

A: $\hat{p} = 803/1794 = 0.448$

	title	binary
1	Dredd 3D	PASS
2	12 Years a Slave	FAIL
3	2 Guns	FAIL
4	42	FAIL
5	47 Ronin	FAIL
6	A Good Day to Die Hard	FAIL
7	About Time	PASS
8	Admission	PASS
9	After Earth	FAIL
10	American Hustle	PASS
11	August: Osage County	PASS
12	Beautiful Creatures	PASS
13	Blue Jasmine	PASS
14	Captain Phillips	FAIL

Step 3: Create a null distribution

How can we assess whether 803 out of 1794 movies passing the Bechdel test ($\hat{p} = 0.448$) is consistent with what we would expect if 50% (or more) movies passed the Bechdel test?

- i.e., is $\hat{p} = 0.448$ a likely value if $\pi = 0.5$?

If 50% of movies passed the Bechdel test, we can model movies passing the as a fair coin flip:

Heads (True) = passed the Bechdel test

Tails (False) = failed to pass the Bechdel test

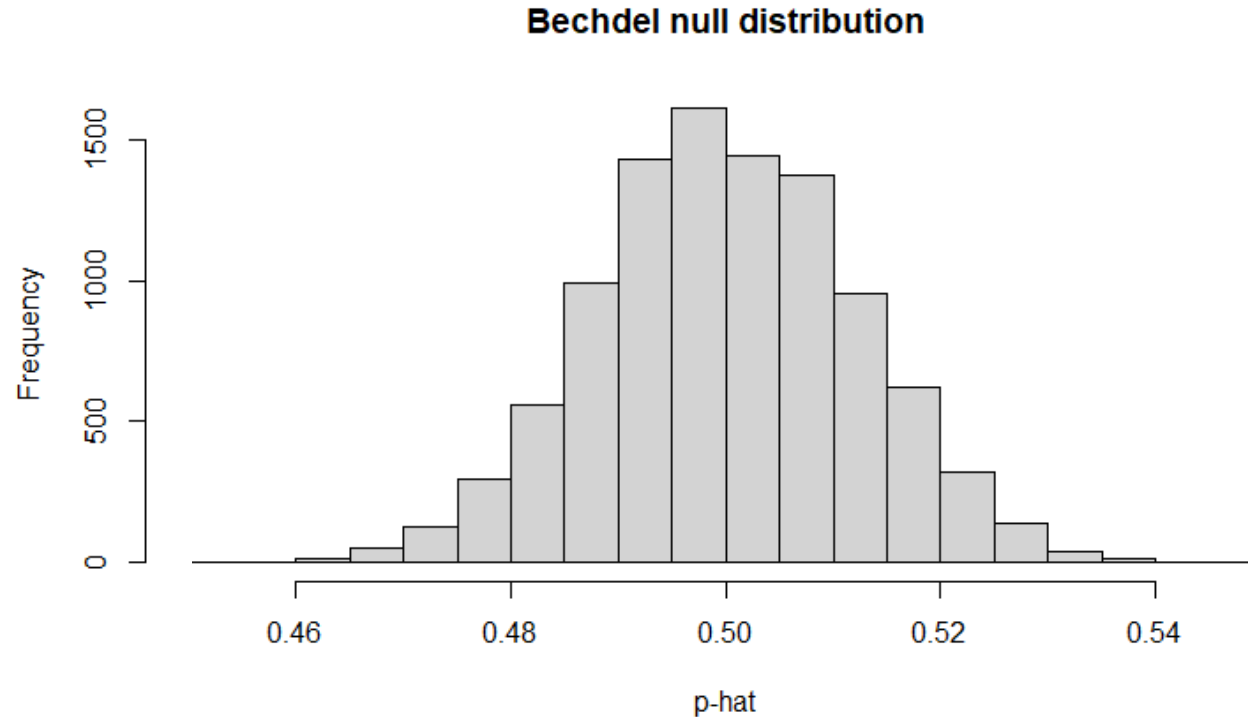
Let's simulate flipping a coin 1794 times and see how many times we get 803 **or fewer** heads

Chance models

To really be sure, how many repetitions of flipping a coin 1794 times should we do?

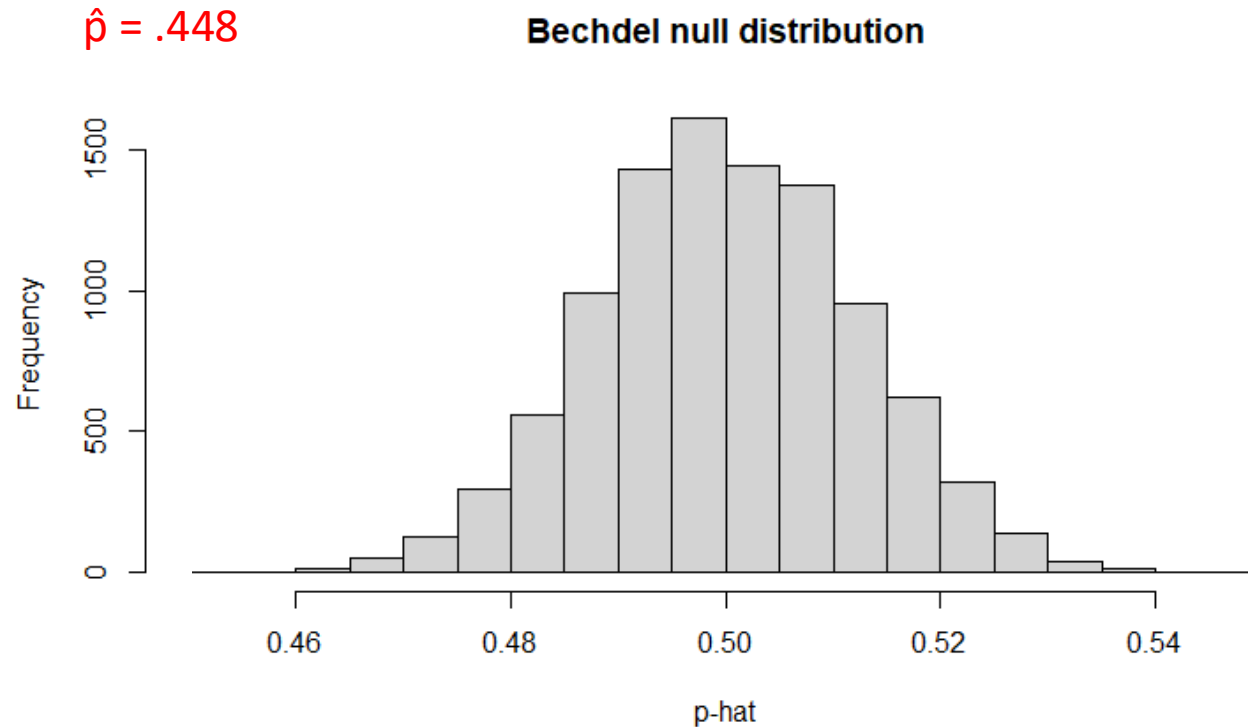
Any ideas how to do this?

Simulating Flipping 1794 coins 10,000 times



Assuming the null hypothesis is true, the distribution of statistics we get is called the **null distribution**

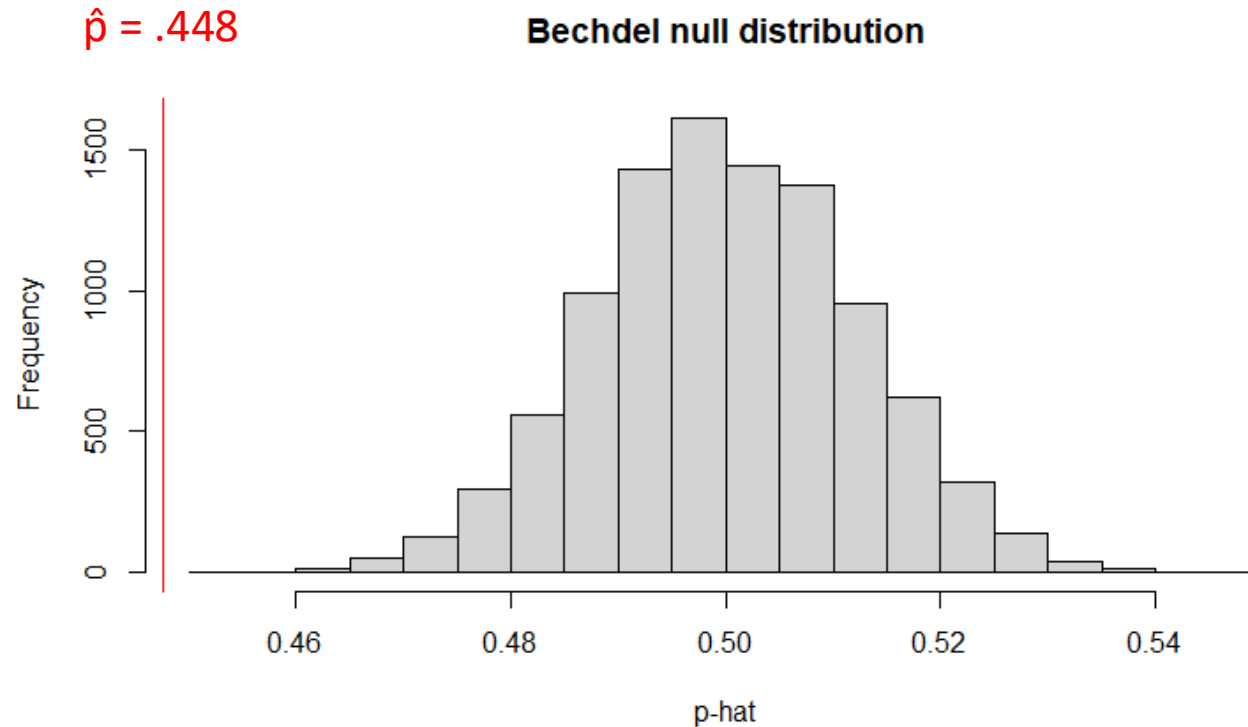
Step 4: calculate the p-value



Q: Is it likely that 50% of movies pass the Bechdel test?

- i.e., is it likely that $\pi = .5$?

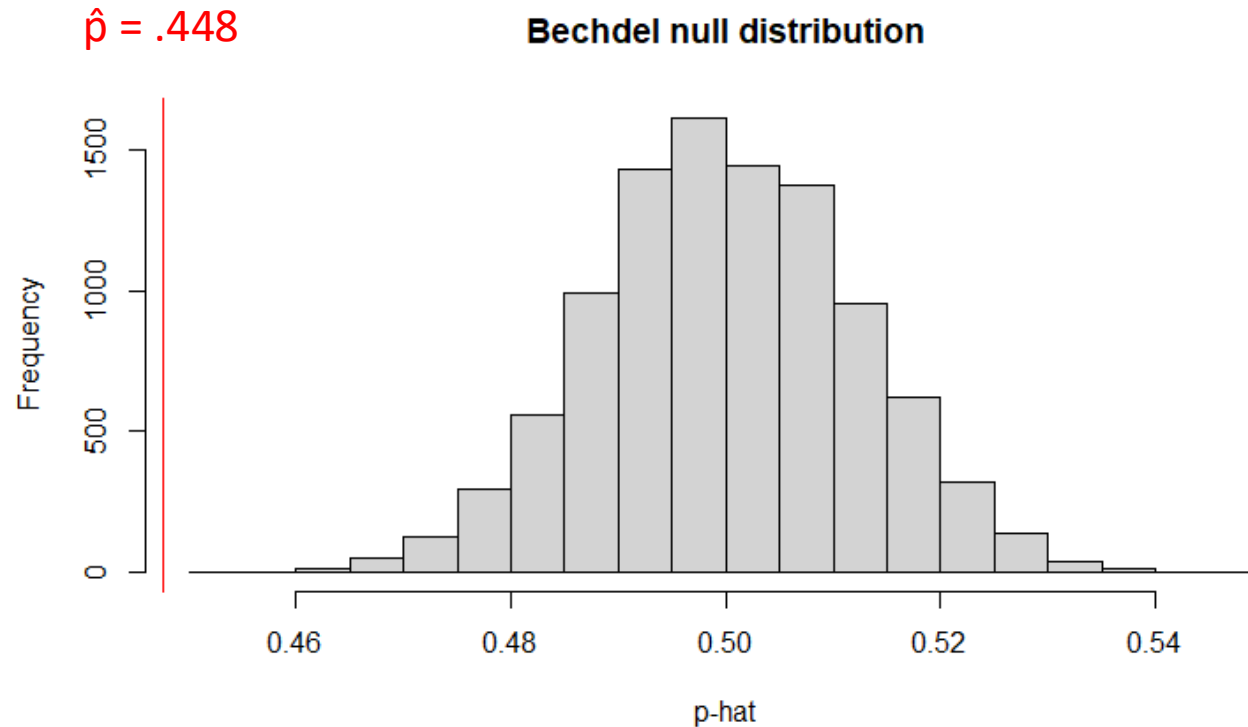
Step 4: calculate the p-value



The **p-value** is the probability we will get a statistic as or more extreme than the observed statistic, if the null hypothesis was true

Q: What is the p-value here? A: the p-value is 0

Step 5: Make a decision

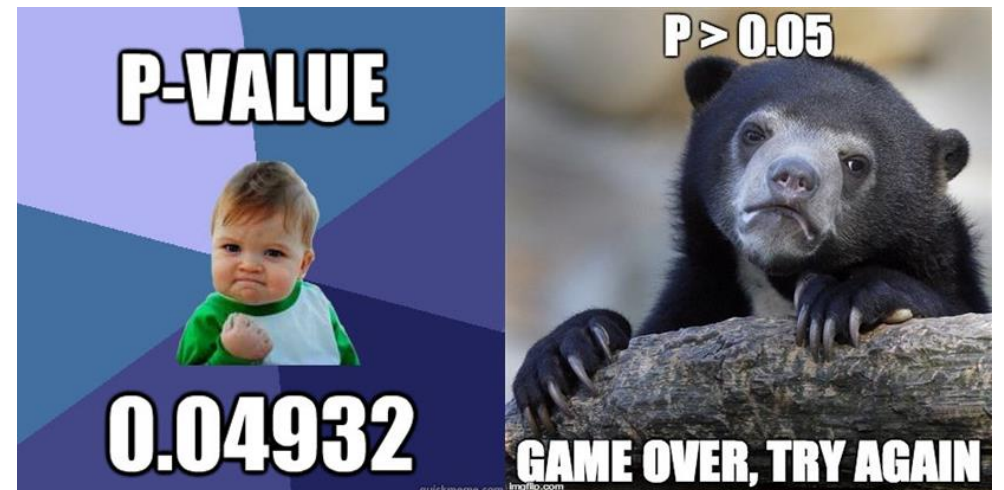


If the observed data is very unlikely if the null hypothesis is true, we can reject the null hypothesis

- i.e., if p-value is very small we can reject the null hypothesis

A few more words

- Why the null hypothesis?
 - "innocent until proven guilty"
 - We will not accept the alternative if the null hypothesis can explain the data
- What is the typical cutoff for p-value?
 - 0.05



Bechdel (hypothesis) test

1. State the null hypothesis and the alternative hypothesis

- 50% of the movies pass the Bechdel test: $H_0: \pi = 0.5$
- Less than 50% of movies pass the: $H_A: \pi < 0.5$

2. Calculate the observed statistic

- 803 out of 1794 movies passed the Bechdel test

3. Create a null distribution that is consistent with the null hypothesis

- i.e., the statistics we expect if 50% of the movies passed the Bechdel test

4. Examine how likely the observed statistic is to come from the null distribution

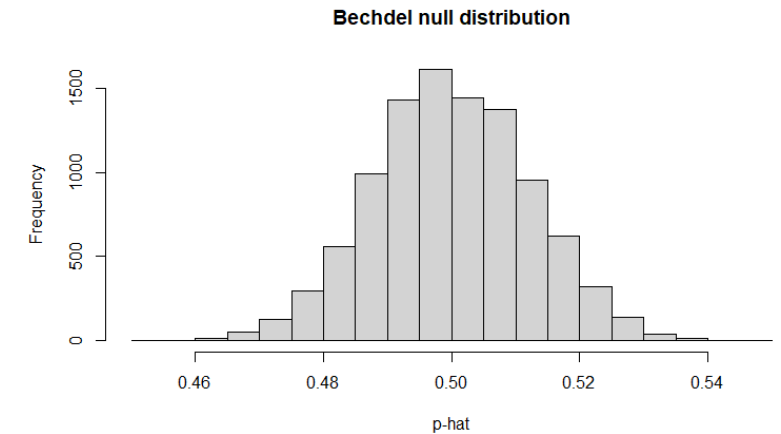
- What is the probability that only 803 of 1794 movies would pass the Bechdel test ($\hat{p} = .448$) if the null hypothesis was true?
- i.e., what is the p-value?

5. Make a judgement

- A small p-value this means that $\pi = .5$ is unlikely, and so it is likely $\pi < .5$
- i.e., we say our results are 'statistically significant'



$$\hat{p} = .448$$



Let's try it in Python!

Another example: sinister lawyers

10% of American population on is left-handed

A study found that out of a random sample of 105 lawyers, 16 were left-handed

Use our 5 steps of hypothesis testing to assess whether the proportion of left-handed lawyers is greater than the proportion on found in the American population

